Individual Contribution Report: Kenneth Kakie

Project Title: Snowflake-Centered Personalized Research Assistant / Legal Demand Assistant **Overall Contribution:** 25%

Personal Contributions by Lab

- **Lab 6: System Integration & Evaluation**
  - Developed the evaluation framework across three distinct scenarios.
  - Implemented the Streamlit-to-Snowflake integration, including secure environment handling and dynamic query execution.
  - Established the initial logging system via `pipeline_logs.csv` to track latency and row-count metrics.
- **Lab 7: Reproducibility & Determinism**
  - Ensured "bit-for-bit" output parity by implementing fixed random seeds across ingestion and embedding stages ($random.seed$, $np.random.seed$, and $torch.manual\_seed$).
  - Authored the `REPRO_AUDIT.md` checklist and the team reproducibility report to verify compliance.
- **Lab 8: QA & Documentation Specialist**
  - Consolidated technical findings from GEPA and QLoRA experiments into the final cohesive report.
  - Performed rigorous testing on the Streamlit "4-config toggle" to verify seamless switching between baseline and adapted models.
- **Lab 9: Monitoring & Structured Logging**
  - Engineered a structured logging system using a custom `RequestContextFilter` to inject query IDs and latency into records.
  - Created the `/metrics` and `/health/snowflake` API endpoints to provide aggregated statistics and connection status.
  - Implemented RAGAS functionality for further evaluation of results.

Evidence of Work

- **GitHub Repository:** https://github.com/BigDataAnalytics-CS5542/snowflake-research-assistant/commits/main/.

- **Key Files & Commits:** `logger.py`, `REPRO_AUDIT.md`, `app.py` (logging logic), and `data/ingestion.py` (seed fixes).

AI Tools Utilized

- **Cursor:** Used for automating refactoring tasks and debugging inconsistencies within the Streamlit UI code.
- **Claude:** Employed to summarize complex evaluation logs into concise tables for group reporting.
- **Antigravity:** Used for initial drafting of technical documentation tailored to the legal domain.

Technical Reflection

My work emphasized that reproducibility is fundamentally rooted in environment isolation and deterministic validation. By automating boilerplate tasks like directory scaffolding and test architecture with agentic AI, I was able to pivot toward high-level architectural decisions, such as structuring JSON artifacts for auditability. I realized that for a research artifact to be credible, it must offer a "one-click" verification path that validates the entire system state.